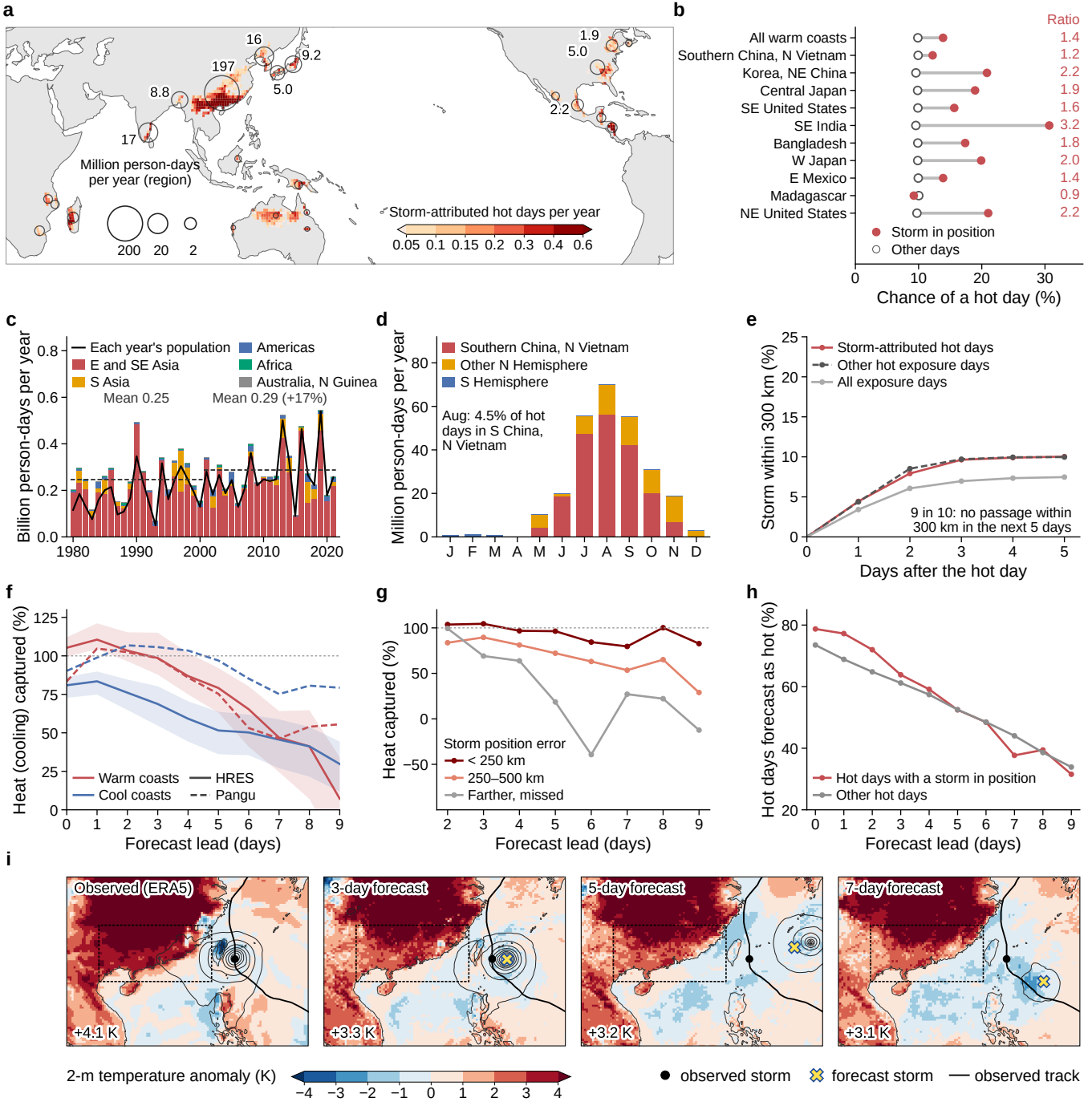

a
16
9.2
197
5.0
8.8
17
1.9
5.0
2.2
Million person-days per year (region)
200 20 2
Storm-attributed hot days per year
0.05 0.1 0.15 0.2 0.3 0.4 0.6
b
Ratio
All warm coasts 1.4
Southern China, N Vietnam 1.2
Korea, NE China 2.2
Central Japan 1.9
SE United States 1.6
SE India 3.2
Bangladesh 1.8
W Japan 2.0
E Mexico 1.4
Madagascar 0.9
NE United States 2.2
Storm in position
Other days
0 10 20 30
Chance of a hot day (%)
c
Each year's population
Americas
E and SE Asia
Africa
S Asia
Australia, N Guinea
Mean 0.25
Mean 0.29 (+17%)
Billion person-days per year
0.0 0.2 0.4 0.6 0.8
1980 1990 2000 2010 2020
d
Southern China, N Vietnam
Other N Hemisphere
S Hemisphere
Aug: 4.5% of hot days in S China, N Vietnam
Million person-days per year
0 20 40 60 80
J F M A M J J A S O N D
e
Storm-attributed hot days
Other hot exposure days
All exposure days
Storm within 300 km (%)
0 5 10 15 20 25
9 in 10: no passage within 300 km in the next 5 days
0 1 2 3 4 5
Days after the hot day
f
Heat (cooling) captured (%)
0 25 50 75 100 125
Warm coasts
Cool coasts
HRES
Pangu
0 1 2 3 4 5 6 7 8 9
Forecast lead (days)
g
Heat captured (%)
−50 0 50 100
Storm position error
< 250 km
250–500 km
Farther, missed
2 3 4 5 6 7 8 9
Forecast lead (days)
h
Hot days forecast as hot (%)
20 40 60 80
Hot days with a storm in position
Other hot days
0 1 2 3 4 5 6 7 8 9
Forecast lead (days)
i
Observed (ERA5)
3-day forecast
5-day forecast
7-day forecast
+4.1 K
+3.3 K
+3.2 K
+3.1 K
2-m temperature anomaly (K)
−4 −3 −2 −1 0 1 2 3 4
observed storm
forecast storm
observed track